Ignite

Adroit Technologies Protocol Driver

Name	Allen Bradley SLC 500-DF1 Serial
DLL	ABSLC500

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Advanced Worx 112 (Pty) Ltd
20 Waterford Office Park
189 Witkoppen Road (c/o Waterford Drive)
Fourways
South Africa

https://adroittech.co.za/

Contents

1. Overview

Driver supports DF1 communication with the Allen Bradley SLC 500 Series controllers, such as MicroLogix 1000, SLC 5/01, SLC 5/02, SLC 5/03, SLC 5/04 Programmable Logic Controllers via RS-232 port.

2. Revisions

2.1. Document Revision

Rev.	Date	Author	Comment	Approval
1.0	14-Dec-2022	J. Duncan	New Format	D. Jordaan
2.0	03-Jan-2023	K. Robinson	Added Device Configuration descriptions for port selection, additional grammar changes.	D. Jordaan
3.0				
4.0				
5.0				
6.0				
7.0				

2.2. DLL Revision

Rev.	Date	Changes
	23-Aug-2000	Initial document for SLC 500.
	25-Jul-2002	Fixed references to old KF driver.
	07-Oct-2002	Setup has PLC model now.
	04-Apr-2003	Application note added, driver is called SLC500 - DF1 now.
	13-Feb-2004	PLC configuration section added in.
9	24-Mar-2011	Zero retries , resulted in no transactions at all, fixed. Added checkbox to purge port on all retries.
11	14-Oct-2011	Exposed PDU length and Intermessage delay controls to setup dialogue.
12	14-Oct-2011	Added markbad tollerance.
13	14-Oct-2011	Format specifier 'U' added, and unreported/possible negative extension read/write bug fixed.
14	14-Oct-2011	Exception injected in rev13, fixed when scanning boolean items.
15	01-Oct-2013	Rev15- Documentation updated.

3. Wiring

3.1. Wiring for SLC 500 RS 232 Port

SLC 500 9-pin		PC 9 pin
1 6		4
2		3
3		2
4		1 6
5		5
7		8
8		7
9	Not used	9

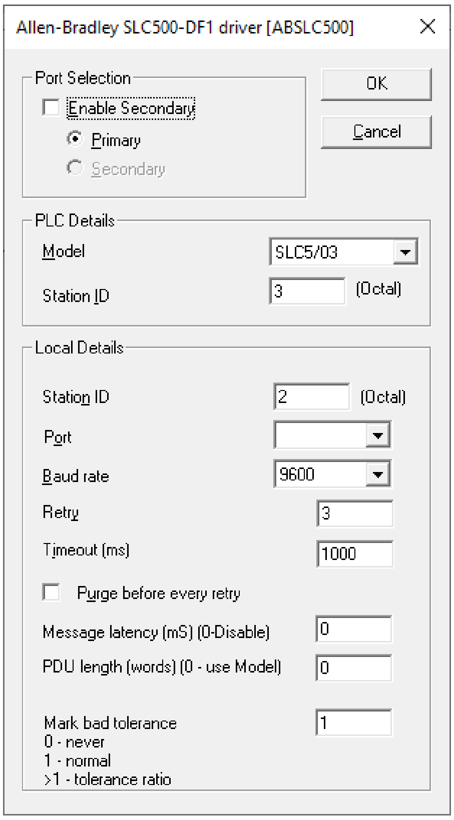
3.2. PLC Configuration

The RS-232 port must be configured for Half-duplex, and CRC checksum. The PLC is the slave. The driver (master) does not support an option for full duplex or BCC. After changing PLC port configuration, you may have to cycle PLC power.

The PLC units may be multi-dropped on the communications line. Create one device agent for each PLC and copy all driver configuration settings except for the destination (PLC) station number. Attach a RS-232/485 converter to each PC and PLC port, and the system will then schedule the polling alternately between devices. The Half-duplex protocol is intended for multi-dropped systems.

Commented [KR1]: Initially said 'on coms line'. Assuming it did not mean COM line?

4. Device Configuration



4.1.1. Port Selection:

Enable Secondary – Turn this option to have the driver maintain 2 channels of communication, where the radio buttons **Primary** and **Secondary** can then be used to toggle between configurations.

4.1.2. PLC Details:

Model – Select the required PLC model (selection will ensure efficient PDU length when building scan tasks.)

Station ID - ID of the destination PLC.

4.1.3. Local Details:

Station ID – The master station's ID.

Port – COM Port to be used.

Baud Rate - Any of the following Baud Rates may be specified: 2400, 4800, 9600, or 19200. The port will be set to **8 data-bits**, **no parity**, **1 stop-bit**.

Retry - The number of times that Adroit will try to gain communication before failing a transaction.

Commented [KR2]: @David Jordaan, does this driver switch to secondary port after the retries have been exhausted?

Time-out - The time that Adroit will wait for a valid response from a PLC before re-trying a transaction.

Checkbox (Purge before every retry) – (Default = unchecked). When checked, every transaction that fails will then purge the COM port before any new attempt at communication. If unchecked the default behaviour will remain – port is only purged after the final failed retry.

Message Latency (mS) – (Default = 0 (off)) A delay before transmitting any request, the value will be clamped to be not more than 5000ms. (Other devices sharing this coms resource will also be delayed)

PDU Length (WORDS) – (Default = 0 (driver model dependant)) The maximum count of data words to send in a single transaction, not including header and footer bytes.

Mark bad tolerance – (Default = 1) 0 means that the driver will never mark any tags bad. 1 means normal operation, and anything greater than 1 means the device will enter a tolerant mode. For example, setting it to 2 and having retries set to 3 will result in the driver only marking bad after the second consecutive total failure (including all retries) which means 2 × (1 + 3) transactions, which means an item will only be marked bad after 9 retry attempts. If the communication recovers before the retries reach 9, then the system will not see any state change.

5. Supported Addresses

Allen Bradley SLC 500 data address types supported by the ABSLC500 Adroit Protocol Driver

Device		Min Range	Max Range	PLC3 Data Type	Boolean	Integer	Real	String
Integer File	**N**	0:0 0:0 U 0:0/0	999:9999 999:9999 U 999:9999/15	Int 16 WORD Bool	- - N 0:0/0	N 0:0 N 0:0 U -	N 0:0 N 0:0 U -	
Decimal	**D**	0:0	999:9999	BCD 16	-	D 0:0	D 0:0	
Floating Point	**F**	0:0	999:9999	AB 32 bit	-	-	F 0:0	
Binary	**B**	0:0/0	999:9999/15	Bool	B 0:0/15	-	-	

Note 1: For Integer and Real type scan-points, specifying a "U" after the address will cause it to report as unsigned 16-bit value. e.g. "N:10:12 U".

Note 2: Some models do not allow certain indexes e.g. SLC5/03 the index starts at 3.

6. Protocol Definition and Considerations

The protocol follows the format of a request message sent by the initiator, the response must be DLE ACK or DLE NAK if the message was not understood. If the message has invalid commands or data, the response is DLE NAK, the response then contains an error code.

The response message contains the data if it is a request or no data for a command with no returned data. The initiator then acknowledges the response or sends a DLE NAK if the response was invalid. *Command codes used are only compatible with the SLC and MicroLogix range controllers.*

6.1.1. Telegram Template

Command ->

DLE	STX	Dest	SRC	CMD	STS	TNS	FNC	DATA	DLE	ETX	CRC
10	02	03	02	0F	00	00 00	A2	**	10	03	XX XX

Response <-

DLE	ACK
10	06

DLE	STX	Dest	SRC	CMD	STS	TNS	DATA	DLE	ETX	CRC
10	02	02	03	4F	00	00 00	**	10	03	XX XX

Acknowledgement->

DLE	ACK
10	06

** See below for format of specific transactions

The CRC is a polynomial 0xA001 compliment of the DEST/SRC bytes up to the end of the DATA bytes and including just the ETX.

Command codes:

0x0F = Command

STS (Status) return codes: (Lo nibble is local code, Hi nibble is remote code)

0x0=no error
0x1=Destination node out of buffer space
0x2=Remote does not acknowledge
0x3=Duplicate token holder detected
0x4=Local port is disconnected

Remote STS codes:
0x0=No error.
0x1=Illegal command or format.
0x2=Host has a problem and will not communicate.
0x3=Remote node is not available.
0x4=Host could not complete function due to hardware fault.
0x5=Addressing problem or memory protect rungs.
0x6=Function disallowed due to command protection selection.
0x7=Processor is in program mode.
0x8=Compatibility mode file missing or comm zone problem.
0x9=Remote node cannot buffer command.
0xA=
0xB=Remote node problem due to download.
0xC=
0xD=
0xE=
0xF=Extended error.

See AB manuals for extended codes

6.1.2. Data Portion of Telegram

This portion contains the data relevant to the command, i.e., for requests it contains an address and a count. For responses it contains only the data.

Sub-command codes:

0xA2=WORD read
0xAA=WORD write

Sub CMD	BYTES	File #	File Type	Element	Sub-element
A2	02	0A	89	0	0

File # is the AB file #
File type is : 89=N file, 8A=Float file, 85=BCD file.
Element is the word #
Sub-element is always 0
The returned data contains only the returned bytes, corresponding in length to the "BYTES" field in the request.

7. General Notes and Troubleshooting

7.1. Contact Details

Email: support@adroit.co.za and drivers@adroit.co.za

Phone: +27 11 658-8100

Web: www.adroit.co.za

7.2. Adroit Technologies Knowledge Base

For additional information, please refer to our Knowledgebase website:

Latest Drivers topics - Features, discussions, tips, tricks, questions, problems and feedback (adroittechnologiesautomation.com)